

VOLTAGE STABILISER USING PIC16F877A

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Voltage stabilisers are used for many appliances in homes, offices and industries. The mains supply suffers from large voltage drops due to losses on the distribution lines *en route*. A voltage stabiliser maintains the voltage to the appliance at the nominal value of around 220 volts even if the input mains fluctuates over a wide range.

Here is the circuit of an automatic voltage stabiliser that can be adapted to any power rating. Its intelligence lies in the program on PIC16F877A—a low-cost microcontroller that is readily available. The circuit, when used with any appliance, will maintain the voltage at around 220V even if the input

mains voltage varies between 180V and 250V.

Here the circuit is shown for a 5A stabiliser. It acts within 100ms to produce a smoothly varying output whenever input mains voltage changes. (Servo stabilisers move a variable contact on a toroidal auto transformer to adjust the output when input goes up and down, which takes seconds.)

The PIC16F877A is an RISC (reduced instruction set computer) microcontroller with 35 instructions, and hence program development with it is rather tough. But, there are good support programs.

Circuit description

The circuit is divided into two sections as it is easy to test them separately: voltage stabiliser controller and volt-

age stabiliser buck-boost. The sections can be joined easily.

Voltage stabiliser controller section. This part of the circuit is built around the PIC microcontroller (see Fig. 1). The 5V supply for the microcontroller is derived from a small iron-core mains step-down transformer having 9.0-9V, 300mA rating, two diodes (1N4007) and a 1000µF capacitor followed by the 7805 regulator.

The ADC input channel 0 at port-A pin 2 of IC2 is used as shown in Fig. 1. Here potentiometer VR1 is connected to +5V and ground through a jumper connection. For the purpose of testing, you can vary VR1 to adjust the voltage from 0 to 5V. The reset circuitry at pin 1 (MCLR) has capacitor C1 and resistor R1. Pin 30 (port-D bit 7) gets a signal (marked as 'D') derived

from the mains supply. Pins 17 and 16 (CCP1 and CCP2) provide the actual pulse-output signal that helps in stabilising the mains power. The signal is a set of equally spaced pulses at about 8 kHz for a 12MHz crystal.

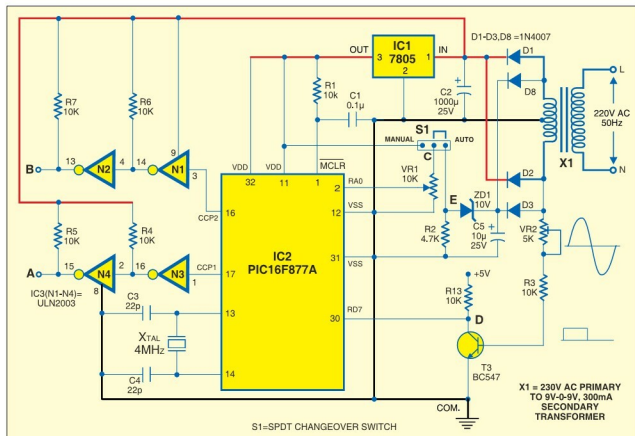


Fig. 1: Circuit of voltage stabiliser controller section